

Chapter 22 — Intraoperative Consultations

- **Operative difficulty rises** with disease, prior surgery, comorbidity and habitus; altered anatomy and unexpected findings raise error and complication rates **however prepared the team**.
- **Definition** — a call from the primary surgeon to a colleague (same or other discipline) for help, prompted by unanticipated additional surgery, iatrogenic injury or unexpected findings; a daily event.
- **Scope:** disease processes, iatrogenic genitourinary injury and trauma; **non-iatrogenic trauma excluded** (trauma chapters).
- **No formal didactics exist;** the precepts are those of everyday surgery, and the challenge is applying judgement and ethics when the situation is unforeseen or inconvenient.
- **Consent gap** — unexpected findings may have been discussed generally, but never the specifics of an unforeseen change of plan.
- **Consultant's role:** advice only → operative help → **complete takeover** when the consultant's skillset best suits the remainder.
- **Constraints:** no prior relationship, history or knowledge of the patient's values; information only from the primary surgeon ± family; the patient is **unconscious and at their most vulnerable**.
- **Injuries recognised postoperatively** carry severe morbidity and far more complex management than those recognised at surgery.
- **Culture:** finding an (especially occult) injury reflects vigilance and integrity; calling for help is courageous and patient-centred; the consultation is **never the time for critique or blame**. Injuries may be careless, unavoidable or **intentional** (oncological cure). Teamwork calms the theatre; being asked is an honour.

Concept — *The consultant inherits a patient who cannot speak, from a colleague under strain; every later rule — stakeholder involvement, deferral of irreversible choices, fact-only documentation — follows from protecting that absent patient without blaming the colleague.*

ETHICAL CONSIDERATIONS

The Patient–Physician (and Family–Physician) Relationship

- **Accepting the consultation enters the consultant into a patient–physician relationship.**
- Often the critical part is **how a life-affecting decision is made**, not the procedure. If stable, **pause to gather information** — primary surgeon, record, family.
- Decisions during surgery require **full disclosure of the reason for consultation** to family or surrogate, so that their choice is informed.
- Such discussions **build trust** (anecdotal evidence), though the patient learns of them afterwards; involving many stakeholders **minimises surprise**.
- Straightforward decisions may proceed without family discussion, followed by a **joint postoperative discussion** (primary surgeon + consultant) with patient and family → transparency and expectations for recovery and quality of life. The consultant may become the follow-up physician → **establish trust early**.

DECISION MAKING

- Mostly straightforward; a minority have **no option free of significant risk**. The consultant makes the **assessment and plan**: alter or keep the plan, **abort**, or add procedures.
- **Stakeholders** (Fig. 22.1, intraoperative decision-making stakeholders): primary surgeon, anaesthesiologist, family or surrogate, urology colleagues — around the consulting urologist. The most important stakeholder, **the patient, cannot participate**, which the consultant must recognise and respect.
- Discussing alternatives often **makes the choice clear**; calling colleagues confirms all options were considered.

ACT NOW VERSUS INVOLVE THE PATIENT

	Low-grade ureteric injury → stent	Complex proximal high-grade ureteric injury, not amenable to ureteroureterostomy
Morbidity	Low	High
Reversibility	Reversible	Often irreversible
Long-term consequence	Unlikely	Substantial impact on quality of life
Action	Proceed without family discussion or waking the patient — extra anaesthesia, second procedure and delay outweigh any benefit	Warrants patient participation → usually temporise and defer
Disclosure	Still mandatory postoperatively	Mandatory

- **Counterweight**: the patient's right to decide is weighed against the harm of delayed repair, a second anaesthetic, a harder reoperation and compounded recovery → in **selected cases, complex repair at consultation** if the harm of deferring outweighs the benefit.
- **Four factors** in temporising versus definitive action: **current patient condition, procedure morbidity, long-term consequences, reversibility**.
- **Rule**: complex reconstruction, with selected exceptions, is **deferred with temporising measures until the patient can take part in shared decision making (SDM)**.

Surrogates.

- May be **absent or unreachable**; the companion may not be the right decision maker. Most hospitals ask for a **pre-designated surrogate** → less guesswork, better knowledge of preferences.
- Surrogates may decide **contrary to the patient's wishes** → strain on the patient-surrogate and patient-physician relationships, especially if recovery is complicated.
- No surrogate, or an emergency → the physician acts in the **patient's best interest**.

Disagreement between surgeons.

- Uncommon. **Default: defer to the primary surgeon** (pre-existing relationship) — **unless deference is potentially harmful**; then the consultant, whose expertise differs fundamentally, acts in the patient's best interest.
- Discussion usually yields consensus; persistent disagreement → **escalate to department chairs or hospital medical director**.

- **Ultimately the consultant is the decision maker:** others inform, but only the urologist has the expertise for the urological problem; act in the patient's perceived best interests, balancing risk and benefit.

Concept — *Every intraoperative choice trades autonomy against the harm of delay: reversible, low-morbidity acts proceed now, while irreversible, life-altering reconstruction waits for the patient unless waiting is itself the greater harm.*

DOCUMENTATION

- Resist **editorialising**; record **observable facts only**.
- **Appropriate:** a urethral stricture as the cause of difficult catheterisation (directly observed); the primary surgeon's fact-based account of mechanism when it changes management — **cautery versus cold-shears ureteric injury** alters the plan and, if directly observed, is documented.
- **Inappropriate:** guessing the cause or assigning blame. Calling an injury "**careless**" or "**avoidable**" in an operative note is itself careless and avoidable.
- The consultant must **scrub in, assess personally and write a separate summary of findings**; relying on the primary surgeon to document is irresponsible and incomplete.
- **Even if no injury is found and no intervention made**, the consultant's primary observations must be recorded — lack of documentation leaves the consultant **vulnerable to legal action**.

Concept — *The note records what the consultant saw, never why it happened: fact protects both patient and consultant, conjecture protects neither.*

KEY POINTS — ETHICS, DECISION MAKING, DOCUMENTATION

- Consultation creates a new patient–physician relationship; full disclosure of its reason is critical.
- Include all relevant stakeholders in difficult intraoperative decisions where possible.
- Temporise or act: weigh patient condition, procedure morbidity, long-term consequences, reversibility.
- Complex reconstruction, with selected exceptions, is deferred with temporising measures until SDM is possible.
- Documentation is fact-based, free of conjecture, and records the consultant's own observations and assessment.

SURGICAL TIME OUT

- The **surgical time out** (preoperative safety checklist) has **no defined role in consultation**: a traditional time out is **not feasible beforehand**, because procedure and supplies are unknown until the field is explored → once information is gathered and family consulted, **review the plan with the whole theatre**.
- **Patient condition**, absent from the standard time out, is added: it varies widely at the time of consultation, and its inclusion confirms the team agrees the patient can tolerate the extra procedure and reaffirms the plan.

TABLE 22.1 SUGGESTED TIME OUT PROCEDURE FOR INTRAOPERATIVE CONSULTATION

Introduction of all staff → patient identification (name, medical record number, date of birth) → **current condition per anaesthesiologist and primary surgeon** → name of procedure → instrumentation reviewed and availability confirmed with theatre staff → fluids for the operative field → estimated blood loss and

expected additional operative time → concerns not yet addressed → **verbal agreement and acknowledgement of the plan by all.**

- **What the table tests:** the standard checklist adapted for a consultation. The distinctive items are the patient's current condition and the added time and blood loss; the trap is timing — it follows assessment, it does not precede it.

Concept — *The consultation time out comes after assessment, not before, because the consultant cannot name the procedure until the field has been seen.*

URETHRAL INJURY

- **Catheter-related male urethral injury in theatre is common** — catheters are passed across disciplines by operators of variable experience.
- **Contributors:** urethral stricture, prior surgery or pelvic radiotherapy, benign prostatic hyperplasia (BPH), operator inexperience.
- Mostly inconsequential; occasionally: **deferral of the planned surgery**, voiding dysfunction, prolonged gross haematuria, acute renal failure, urethral stricture, prolonged catheterisation.
- Adds **cost and length of stay**; catheter duration drives bacteriuria and **catheter-associated urinary tract infection (CAUTI)**, affecting reimbursement. Severe injuries rarely need surgery.
- **History is central** — prior cystoscopic or abdominal surgery, prostate cancer, prostatectomy, radiotherapy, anticoagulation, planned operation → dictates catheter choice, cystoscopic placement, and suprapubic risk.
- **Female urethral injury** occurs in pelvic reconstructive surgery, sometimes with concomitant bladder or ureteric injury → markedly more complex repair.

Incidence

- Theatre incidence **unknown**; in hospitalised patients **3.2–6.7 injuries per 1000 men catheterised**.
- Most men with an iatrogenic urethral injury suffer a complication of **Clavien–Dindo grade 2 or higher**.

Mechanisms

- **Commonest:** benign prostatic enlargement plus an inexperienced operator.
- Two mechanisms: **balloon inflation within the urethra**, and **false passage** created by forcing the catheter tip. A tortuous course or stricture increases false passage.

Male urethra	Female urethra
Long, tortuous , stricture-prone → catheter injury common	Short, straight, stricture rare → catheter injury far less common
Balloon inflation in urethra; false passage	Pelvic reconstructive surgery: sling placement, prolapse repair

Diagnosis

- **Male:** failed catheterisation ± blood per urethra or in catheter; cystoscopy confirms and gauges extent. **Retrograde urethrography** possible but more valuable in trauma (disruption needing urethroplasty likelier).
- **Female:** mostly by **cystoscopy**, often done precisely for this purpose.
 - **Obvious:** trocar or mesh visible within the urethra.

- **Subtle: tethering of urethra or bladder neck** (trocar or mesh through the wall, lumen not breached); catheter seen through an inadvertent urethrotomy during dissection between anterior vagina and posterior urethra.

Management

- **Male catheter injury:** specific catheter choice → **one blind attempt** by the urologist (as in trauma) → if it fails, **cystoscopy, guidewire into bladder**, catheter over the wire → if stricture or injury precludes safe placement, **percutaneous or open suprapubic catheter**.
- **Duration of catheterisation** depends on extent of injury, need for concomitant urethral dilatation, and postoperative condition.
- **Non-catheter urethral injury is rare** — the urethra lies outside most operative fields → mainly **transvaginal or transperineal reconstruction**.
- **Female repair:** vaginal approach carries **fistula risk** → interpose **peritoneum, omentum or labial fat** between anterior vaginal wall and bladder or urethra when fistula risk is anticipated.

Concept — *In men the urethra is injured by the catheter, in women by the trocar or dissection; anatomy predicts who is injured, and one blind attempt followed by vision is the universal rule.*

BLADDER INJURY

- Evaluation and management overlap heavily with bladder trauma (Chapter 125); this section addresses **iatrogenic** injury.

Incidence

- **The bladder is the most commonly injured organ in pelvic surgery.**
- The pelvis is a **fixed space** → hard exposure and retraction, little room. **Distortion** (tumour, prior surgery, radiotherapy, prolapse) and vaginal-approach reconstruction raise risk.

TABLE 22.2 INCIDENCE OF BLADDER INJURY IN COMMON PELVIC SURGERIES

Category	Procedure	Incidence
Obstetric	Caesarean section	0.27–0.3%
Hysterectomy	Laparoscopic total / abdominal / vaginal / laparoscopic supracervical	0.75% / 0.58% / 0.51% / 0.29%
	Laparoscopic-assisted vaginal	Not stated
Continence surgery	Transvaginal (retropubic) tape / transobturator tape	3–9% / 0.5%
Hernia	Inguinal (open) / laparoscopic	0.08–0.3% / 1.6%

- **Pattern:** all are low except retropubic tape, where the trocar traverses the space beside the bladder — hence routine cystoscopy. **Trap:** the laparoscopic variant carries the higher rate for both hernia and hysterectomy.

Mechanisms

- **Laceration commonest; devascularisation** rarer.
- Risk is highest operating **within** the bladder: **transurethral resection of bladder tumour (TURBT)** is the commonest and most under-reported source. TURBT and biopsy are **intentional injuries** that usually heal, with or without catheter decompression.

- Management set by haematuria, postoperative anticoagulation, perforation size, anticipated continuous bladder irrigation, **abdominal distension during endoscopy** — usually the primary urologist's own problem, not a consultation.

Extraperitoneal endoscopic perforation	Intraperitoneal endoscopic perforation
Most → prolonged catheter drainage	Most → open repair

TABLE 22.3 GRADING SYSTEM FOR CLASSIFYING BLADDER INJURY

Grade	Type	Description
I	Haematoma / laceration	Contusion, intramural haematoma / partial-thickness laceration
II	Laceration	Extraperitoneal wall laceration <2 cm
III	Laceration	Extraperitoneal ≥2 cm or intraperitoneal <2 cm
IV	Laceration	Intraperitoneal ≥2 cm
V	Laceration	Extends into trigone (bladder neck or ureteric orifice)
VI	—	Not salvageable

Moore EE, Cogbill TH, Jurkovich GJ, et al. Organ injury scaling III. J Trauma 1992.

- **Pattern:** grade rises with size and with the peritoneal compartment. **Trap:** an intraperitoneal tear under 2 cm is grade III, not II; trigonal extension is grade V whatever the size.

Diagnosis and Recognition

- **High index of suspicion.** Indicators: visible laceration, **visible Foley catheter**, extravasation of fluid (urine) into the field, blood (and gas at laparoscopy) in the catheter bag.
- Suspicion without a visible defect: **instil fluid ± methylene blue** via the catheter; **contrast cystography** if fluoroscopy available; **cystoscopy** — now **standard of care after urethral sling surgery and hysterectomy**.
- Once bladder injury is found, **exclude concomitant ureteric or orifice injury**, especially with trigonal injury.

Management

- **Repair sequence:** debride devascularised urothelium or wall → close in **one or more layers, absorbable suture** → **test by filling** (allows extra sutures and exposes missed injuries) → **urethral catheter plus intra-abdominal drain** → drainage **7–14 days**.
- **No routine suprapubic tube** alongside a urethral catheter — no supporting data.
- **Postoperative cystography selective** (mostly negative): doubtful integrity, complex repair or fistula concern — confirms no leak before catheter removal.
- Injury **suspected but not seen** → **cystotomy** to find it.
- **Trigonal injury:** establish ureteric integrity; anterior vaginal wall may be involved → fistula risk. **When in doubt, reimplant the ureter** — an unrepaired ureteric injury has serious consequences.

Concept — Every bladder repair answers three questions — *is it closed (fill test), is it drained (catheter and drain), are the ureters intact* — and the third is the one missed at the trigone.

URETERAL INJURY

Incidence

- **Iatrogenic injury is the commonest cause of ureteric injury (75–80%),** with every approach; rarely after spinal surgery (seldom recognised intraoperatively). Fixed pelvis, complex anatomy, prior surgery and radiotherapy displace and obscure the ureter.
- **Over 90% involve the distal third.**
- **Gynaecological surgery causes the majority (>50%);** colorectal, vascular and urological also. Some series now find **endoscopy** leading, but **endoscopic avulsion is now rare** with modern equipment. Minimally invasive surgery is thought to increase injury.

TABLE 22.4 INCIDENCE OF URETERAL INJURY BY TYPE OF SURGERY

Surgery	Incidence
Transvaginal hysterectomy	0–1%
Transabdominal hysterectomy	0.35–30%
Colorectal surgery	0–1.9%
Ureteroscopy	0.04–4.1%

- **Pattern:** abdominal hysterectomy has the widest and highest range, vaginal the lowest. **Trap:** ureteroscopic injury is frequent overall, but avulsion is rare.

Mechanisms

- **Laceration** (partial or complete), **ligation** (partial or complete, with kinking or occlusion), resection, crush, thermal injury, abrasion, avulsion, devascularisation. **Mechanism determines grade and management.**

TABLE 22.5 GRADING SYSTEM FOR CLASSIFYING URETERAL INJURY

Grade	Type	Description
I	Haematoma	Contusion or haematoma without devascularisation
II	Laceration	<50% transection
III	Laceration	≥50% transection
IV	Laceration	Complete transection, <2 cm devascularisation
V	Laceration	Complete transection, >2 cm devascularisation
VI	—	Not salvageable

Best CD, Petrone P, Buscarini M, et al. Validation of the American Association for the Surgery of Trauma Organ Injury Scale for ureteral injury. J Urol 2005.

- **Pattern:** grades II–III turn on fraction of circumference, IV–V on **length devascularised**. **Trap:** IV and V are both complete transections — the 2 cm of devascularised ureter separates them.

Diagnosis and Recognition

- **Most iatrogenic ureteric injuries go unnoticed:** only about a third are recognised at open surgery, fewer still at laparoscopy.
- **Missed injury** → pain, infection, azotaemia, fistula, reoperation; **immediate recognition and repair** gives better outcomes and fewer complications.
- **Signs:** extravasation of fluid (urine), hydroureteronephrosis, a visible pre-placed ureteric catheter; subtle — **wall discolouration, absent capillary refill.**

- **Haematuria raises suspicion, but its absence does not exclude significant injury.**

INTRAOPERATIVE ASSESSMENT OF THE URETER

Modality	Role	Caveat
Cystoscopy + retrograde pyelography	Gold standard: most accurate for location and degree; a normal study can obviate further dissection; ideal during endoscopy	Often impossible in open or laparoscopic positioning
Cystoscopy alone (efflux)	Screening	Efflux persists with partial transection or ligation → add retrograde study when suspicion high
Ureterscopy; ureteric catheter or stent passage	Unobstructed passage to renal pelvis → no significant ligation or laceration	Same positioning limits
Intraoperative excretory urography	When retrograde access impossible	Abnormal or non-diagnostic → explore
One-shot intravenous pyelogram (IVP)	—	Inadequate: negative in up to 60% of injuries (European Association of Urology [EAU] guidelines)
Excretory dye (indigo carmine, methylene blue, sodium fluorescein) — intravenous, via ureteric catheter, or needle into pelvis or ureter	Direct instillation is faster	Dye in the field = higher-grade injury → explore
Direct inspection after meticulous dissection	When retrograde study unavailable; look for laceration, ligation, contusion (bruising, haematoma), devascularisation (dusky, no bleeding)	Open is gold standard; laparoscopic or robotic per consultant's skill — modality set by preference and proficiency

Management

- Set by location, mechanism, **patient and consultant's expertise**; goals: **preserve renal function, minimise morbidity.**
- **American Urological Association (AUA):** repair traumatic ureteric injury **at exploration in stable patients (grade C)**; extrapolated to acute iatrogenic injury.
- **Stable:** establish planned operation, indication, history, suspected mechanism; review imaging for planning.
- **Unstable patient → temporary diversion** and delayed definitive repair: **ligate ureter + percutaneous nephrostomy**, or ureterostomy. Also if the consultant lacks the skill — though seek help to spare repeat procedures; **first do no harm.**

Principles of Repair

Same at every level: **debride devascularised tissue → spatulated, tension-free, mucosa-to-mucosa anastomosis with absorbable suture → watertight closure → indwelling stent** (keeps lumen patent, directs urine across rather than out of the anastomosis) → **external drain** → interpose omentum or peritonealise where possible to aid healing and prevent fistula.

REPAIR BY GRADE AND LEVEL

Injury	Repair	Key point
Grade I–II (ligation, cold laceration, crush, minor thermal)	Remove clip or suture + ureteric stent (or nephrostomy)	Stent preferred over nephrostomy — less stricture. Ligation is the commonest gynaecological ureteric injury
Grade III–V distal	Ureteroneocystostomy → psoas hitch → Boari or advancement flap	Distal ureteroureterostomy has compromised vascularity
Grade III–V mid and proximal (above the iliacs)	Primary repair over a stent (AUA grade C; EAU)	Ureteroureterostomy best for small ligation and crush injuries not manageable by stent alone
Complex (grade V)	Temporise and defer	Needs SDM
Ureteroscopic avulsion	Mid/proximal: immediate ureteroureterostomy ; distal: reimplant; proximal: ureterocalicostomy	Re-establish continuity as for grade IV laceration

Grade I and II Injuries

- Stent also for contusion or small laceration. **Stent 2–6 weeks**; follow-up evidence lacking → image upper tract (**ultrasound or computed tomography [CT]**) after removal; **renal scan** if indeterminate.

Grade III to V Distal Ureteral Injuries

Refluxing reimplantation	Non-refluxing reimplantation
Preferred acutely — efficient, lower anastomotic obstruction	No clear advantage in adults
• Insufficient length → psoas hitch: ipsilateral superior detrusor sutured to ipsilateral psoas tendon; developing the space of Retzius and ligating the contralateral bladder pedicle mobilise further → still short: tubularised bladder flap (Boari) or bladder advancement flap (technique: Chapters 53 and 118). • After reimplantation: urethral catheter 1–2 weeks + perivesical drain → selective cystography (complex or doubtful repair) → stent stays after catheter removal, total 4–6 weeks → image after all drains out to exclude stricture.	

Grade III to V Mid and Proximal Ureteral Injury

- Above the iliacs: **primary repair over a stent where possible** (see table).

Grade V Injuries

- Complex proximal repairs — **ureterocalicostomy, transureteroureterostomy, ileal ureter, autotransplantation** — carry significant risk and long-term effects on health and quality of life → **strong consideration to temporise and defer**.
- **Appendiceal onlay and buccal mucosal grafts** (for ureteric stricture) may offer less morbidity, but are **unexplored at intraoperative consultation**.
- Need expertise and SDM; an informed family decision here is arguably impossible → **temporise**.

Ureteroscopic Avulsion Injuries

- **Rare: 0.04–0.9%**. Proximal to mid ureter **most prone**; usually **excessive force on a basket around an impacted stone**.
- Primary goal: **re-establish continuity** (options in table above).

Ureteral Catheters

- Requests (before colorectal or gynaecological surgery) vary by institution; for anticipated distortion — **diverticulitis, large pelvic mass, prior radiotherapy or surgery.**
- **Prophylactic catheters do not reduce the risk of ureteric injury, but aid recognition** of injury when it occurs → less morbidity.
- **Risks:** iatrogenic ureteric injury; added cost; **more postoperative acute kidney injury (AKI)**, greater with **bilateral than unilateral** catheters — catheter-related ureteric oedema the likely cause.
- Almost always anticipated → **counsel preoperatively by a urologist**; the primary surgeon's ethical duty is to reserve intraoperative consultation for **unanticipated** findings.

Concept — *Recognise, then let the level choose the repair: distal injuries go to the bladder, mid and proximal injuries are joined end to end, and anything needing irreversible reconstruction waits for the patient.*

RENAL INJURY

Incidence

- **Mostly traumatic** → renal consultation rare; iatrogenic injury rare but morbid.

Mechanisms

- Trauma: blunt and penetrating. Iatrogenic: **haemorrhage** from vascular or parenchymal injury — percutaneous nephrolithotomy (PCNL), renal biopsy, endopyelotomy, open abdominal and endovascular surgery.
- **PCNL is now the commonest cause of renovascular injury (0.9–3.0%).**

Diagnosis and Recognition

- Mostly recognised **postoperatively**: gross haematuria, acute blood-loss anaemia, flank pain or haematoma, **haemodynamic instability.**
- Most → **conservative or selective embolisation**; acute renal bleeding rarely reaches urological consultation (renal trauma: Chapter 125).

Management

- Renal trauma trend: **conservative even for high grades. AUA (2017, grade B): immediate intervention (surgery or angioembolisation) only if haemodynamically unstable or only transiently responsive to resuscitation.**
- Iatrogenic injuries are mostly minor → supportive care.

Venous PCNL bleeding (usual)	Arterial bleeding
Catheter tamponade, clamp nephrostomy, mannitol, upsized nephrostomy → no exploration	Selective angioembolisation ; open exploration generally not indicated

- **Angioembolisation versus open exploration** (partial or total nephrectomy): less parenchymal loss and morbidity.

Concept — *The kidney is rarely a theatre consultation because its iatrogenic injuries declare themselves after surgery and stop with tamponade or embolisation; exploration is last because it costs nephrons.*

KEY POINTS — BLADDER AND URETER

- After bladder injury, exclude ureteric or orifice injury.

- Bladder injury suspected but not seen → cystotomy to locate it.
- Early recognition of ureteric injury matters; failure carries significant morbidity.
- Ureteric injury suspected but not seen → imaging and/or endoscopic assessment.
- Ureteric catheter placement belongs to the preoperative setting and selected cases.

PLACENTAL CASES

- Abnormal placental attachment classified by depth: **accreta** → **increta** → **percreta**, with increasing invasion and complexity; "placenta accreta" often denotes the whole spectrum. High risk of haemorrhage and morbidity.
- **Incidence rising**, likely from increasing **caesarean delivery** (a known risk factor). **Most accreta occurs with placenta praevia** (low-lying placenta partly or wholly covering the cervix) → suspect and evaluate accreta in every praevia.
- **Antenatal diagnosis** enables planning. **Ultrasound** is commonest; sensitivity data conflict — high in high-volume centres, yet most cases still go undetected before delivery → high clinical suspicion needed.
- **Magnetic resonance imaging (MRI)**: unclear added accuracy, no clear change in management or outcome; may define **depth of invasion in percreta**. Imaging studies are biased (expert centres, high-risk patients).
- **Management: planned early caesarean at 34–38 weeks** (avoids emergency surgery in labour); hysterectomy at the same time avoids dissecting the placenta; **controlled caesarean hysterectomy = gold standard** — less morbidity than attempted placental removal with uterine preservation. Other options: **internal iliac artery embolisation**, postoperative **methotrexate**.
- **Multidisciplinary team including urology**, because percreta can invade bladder and ureter; **commonest invasion site: posterior bladder**.

Urologist's roles.

- **Antenatal**: evaluate suspected bladder percreta. Commonest presentation is **bleeding — vaginal bleeding far more often than haematuria**.
- **Cystoscopy is insensitive** (mucosa may be normal despite muscularis invasion) but **still warranted** for evaluation and planning. **Never biopsy** suspected percreta with cystoscopic abnormality — risk of **exsanguinating haematuria**.
- **Intraoperative**: consultation when accreta was unknown or antenatal care absent. Cystotomy may be intentional or incidental → **complete resection of placental tissue, two-layer closure**, consider omental interposition against fistula → urethral catheter on dependent drainage ± pelvic drain → cystography before catheter removal per repair quality and surgeon preference.
- **Subtotal hysterectomy** with invasive tissue left may be chosen when removal would cost anatomy or function (e.g., defunctionalised bladder capacity, need for ileal ureter); methotrexate has been used to shrink residual tissue — but **haemorrhagic shock** has occurred with retained placenta → **standard of care remains removal of all placental products**.
- **Ureteric injury or involvement in up to 7%**. Ureteric catheters **may reduce ureteric injury and early morbidity in placental cases**, elective and emergency.
- Suspected ureteric injury → manage as above: do what is best for the patient and within the consultant's skillset.

Prophylactic ureteric catheters — general pelvic surgery

Do not reduce injury; aid recognition; increase AKI and cost

Prophylactic ureteric catheters — placental abnormality

May reduce injury and early morbidity, elective and emergency

Concept — *Percreta invades the bladder like a tumour but bleeds like a placenta; hence no biopsy, complete excision, two-layer closure and ureteric protection.*

CONSULTS FROM UROLOGY COLLEAGUES

- Overlapping skillsets → the call implies **greater complexity or a hard decision**; or the on-call urologist lacks the skillset for another discipline's consult; or simply to **reaffirm a decision**.
- A consult from within urology is a **particularly high honour and sign of respect**.